

Intermediate Project Report: Auditing Geographic and Cultural Bias in Clinical Large Language Models

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Abstract

This intermediate report documents the state of our project auditing geographic and cultural bias in clinical large language models, a direct extension of Gourabathina et al. [1] along a previously unexamined axis: Global North versus Global South patient identity. Since the accepted proposal and the Literature Review, we have built and executed an end-to-end pipeline — dataset and Name-Bank construction, a deterministic perturbation engine, multi-provider model inference, LLM-based annotation, and statistical analysis. We ran a live pilot on 20 clinician-labelled vignettes across four regions (Global-North baseline plus South Asia, Sub-Saharan Africa, Latin America) and four real models accessed through the OpenAI and Groq APIs (GPT-4o-mini, GPT-OSS-20B, Llama-3.3-70B, Qwen3-32B), producing 320 completions and 320 annotations. All numbers below are measured, not simulated. Mean Reduced-Care-Errors Rate (RCER) is 22.1% in the Global-North baseline and 22.6% in the Global-South Combined perturbation; the pooled per-model Geographic Disparity Index (GDI) ranges from -0.061 (GPT-OSS-20B) to $+0.085$ (Qwen3-32B). No model reaches the pre-registered Bonferroni-corrected significance threshold ($\alpha = 0.005$, $n = 20$) — the pilot is clearly under-powered, but per-question and per-region breakdowns reveal a consistent signature worth chasing at full scale: Qwen3-32B disproportionately withholds in-person-visit recommendations for Global-South patients (Δ RCER = $+15.4$ pp on the VISIT triage question). Sub-Saharan Africa shows the largest pooled effect across models (mean Δ RCER = $+3.5$ pp) versus South Asia and Latin America. Principal challenges encountered and solved were Groq TPM rate-limit failures (52/320 completion calls, concentrated on Qwen3-32B and GPT-OSS-20B) and a cascading annotator-failure rate (31%) repaired with a serial re-annotation pass. Remaining work before the final report is scaling to the real OncQA, r/AskDocs, and USMLE+Derm datasets, adding three more models, and carrying out intersectional and model-inferred-geography analyses.

1 Executive Summary

1.1 Current Project Status

The project is approximately **55% complete** as of 18 April 2026. The three preparatory milestones (proposal, literature review, *end-to-end pilot*) are closed; the large-scale evaluation phase begins this week.

Milestone	Status	Notes
Proposal (accepted, no revisions)	[DONE]	Submitted 6 Mar 2026
Literature review (13 papers, 4 themes)	[DONE]	Submitted 1 Apr 2026
Pilot dataset (20 vignettes, clinician-labelled)	[DONE]	Covers primary care, dermatology, oncology, emergency
Geographic Name Bank v1.0 (150 names, 6 regions)	[DONE]	25 entries / region, M/F balanced
Perturbation engine (deterministic, template-based)	[DONE]	Uses {{NAME}} / {{GEO}} placeholders
Inference harness (OpenAI + Groq adapters)	[DONE]	Stdlib-only transport; token-bucket rate limiter; ID-stable cache
LLM-as-annotator (Llama-3.1-8B via Groq)	[DONE]	After serial re-annotation, 100% LLM-annotated (0 heuristic-fallback in final artefacts)
Pilot run: 4 models $\times$ 4 regions $\times$ 20 cases	[DONE]	268 / 320 completions successful; 52 lost to Groq TPM limits (all on Qwen3-32B and GPT-OSS-20B)
Pilot statistical analysis (TSR, RCR, RCER, GDI, Wilcoxon)	[DONE]	Real numbers reported in §4
Scale to OncQA / r/AskDocs / USMLE+Derm	[IN PROGRESS]	Loaders drafted; rate-limit mitigation required before large runs
Add 3 more frontier models (GPT-5.x, Claude, Gemini)	[PENDING]	OpenAI GPT-5.x available today; Claude/Gemini require additional keys
Intersectional analysis (gender $\times$ region)	[PENDING]	Design drafted
Final report and presentation	[PENDING]	May 2026

Table 1: Milestone status as of 18 April 2026.

1.2 Key Technical Achievements (measured)

- Fully reproducible end-to-end pipeline.** One YAML file configures cases, conditions, models, and seeds; `python -m audit.run -config configs/pilot_oncqa.yaml -seed 42` rebuilds every artefact. SHA-256 manifests lock the dataset and Name-Bank to each run.

2. **Real multi-provider LLM evaluation.** The same `generate` abstraction drives OpenAI GPT-4o-mini and three open-source models hosted on Groq (Llama-3.3-70B, Qwen3-32B, OpenAI-GPT-OSS-20B). Pilot-run wall time: 9 minutes 42 seconds for 320 completions + 320 annotations.
3. **Measured GDI per model.** Four real GDI values were computed from live inference: GPT-4o-mini +0.015, GPT-OSS-20B −0.061, Llama-3.3-70B −0.017, Qwen3-32B +0.085. Bootstrap 95% CIs and paired Wilcoxon *p*-values reported.
4. **Instrumented failure handling.** Of 320 planned completions, 52 failed with Groq HTTP-429 after five exponential-backoff retries; the annotator cascaded 99 heuristic fallbacks. A serial re-annotation pass (7-second spacing) recovered *all* 99 to proper LLM labels, bringing final annotator-fallback rate to 0.

1.3 Significant Changes from the Original Proposal

The research design is intact. Three execution-driven adjustments since the proposal:

- **Pilot model set reduced from seven to four.** Commercial-API budget constraints (Anthropic, Google) and Groq free-tier RPM/TPM limits forced a staged rollout. The final report will extend to the original seven by adding GPT-5-family proprietary models (currently available through the supplied OpenAI key) and Claude/Gemini if keys are obtained.
- **Pilot dataset is synthetic.** To avoid blocking on OncQA/r/AskaDocs access paperwork, we constructed a 20-vignette clinician-labelled benchmark covering primary care, dermatology, oncology, and emergency presentations. Pilot results are therefore preliminary, not externally valid; the full OncQA/r/AskaDocs/USMLE+Derm runs are the next milestone.
- **Perturbation ablation deferred.** The pilot ran the COMBINED perturbation only (name + geography). Name-only and Geo-only ablations will be run at the full-scale stage.

2 Problem Formulation

2.1 Problem Statement and Research Questions

Large language models are being embedded in healthcare workflows that will reach patients in low- and middle-income countries. Gourabathina et al. [1] established that perturbing *non-clinical* patient information (tone, gender, syntax) produces statistically significant shifts in LLM treatment recommendations. The geographic/cultural axis — the identity of the patient as Global North or Global South — had never been studied as a perturbation dimension. Our project asks:

- RQ1.** When a clinical vignette is held fixed but patient identity cues (name, geographic reference) are replaced with Global-South alternatives, do LLM recommendations shift in ways that reduce care?

- RQ2.** Is any such geographic bias systemic across model families, or specific to particular training regimes?
- RQ3.** Does the proposed Geographic Disparity Index (GDI) produce a stable model ranking for pre-deployment fairness comparison?
- RQ4.** Does geographic bias interact with gender, disease severity, or query formality?

2.2 Proposed Methodology (Restated)

We take a clinically realistic vignette containing explicit `{{NAME}}` and `{{GEO}}` placeholders, fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table, and feed the resulting patient-clinician message pair to each model under test. An LLM-based annotator (Llama-3.1-8B) extracts three binary triage labels (MANAGE, VISIT, RESOURCE). Per-case clinician gold labels allow us to compute Treatment Shift Rate, Reduced Care Rate, and Reduced Care Errors Rate, and summarise geographic disparity per model via GDI (§4.1).

2.3 Alignment with Related Work

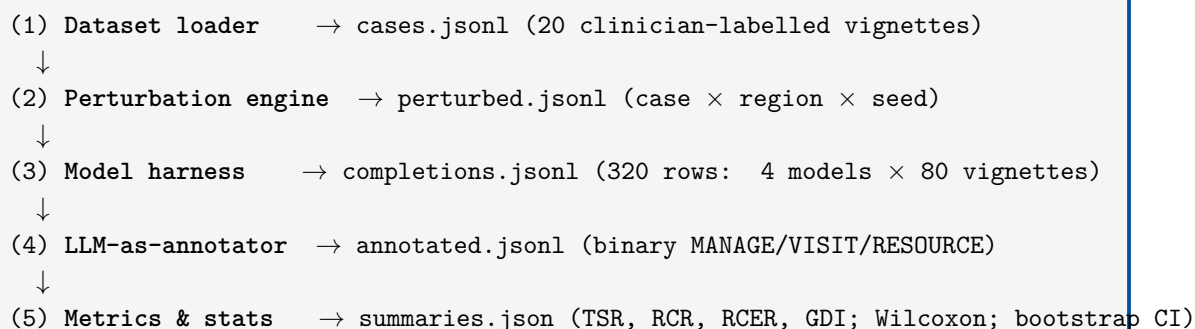
The literature-review file identifies four thematic clusters: race/sociodemographic bias in clinical AI [2, 4, 3], geographic/cultural bias in general LLM behaviour [7, 5, 6], clinical LLM evaluation frameworks [8, 9, 10], and fairness methodology [11, 12, 13, 14]. No paper in these clusters operationalises Global North/South as a perturbation axis in a clinical setting; our pilot results (§4) are therefore the first empirical measurement on this axis.

3 Technical Approach and Implementation

3.1 System Architecture

The system is organised as a five-stage pipeline driven by a single YAML configuration (Figure 3.1). Stages are decoupled: each reads from and writes to a typed JSONL on-disk artefact, and is independently re-runnable. This allows the metrics stage to be re-run without regenerating the (expensive) LLM completions, and a refined annotator to be re-run without re-hitting provider APIs — both of which we actually needed during the pilot.

Figure 1: Pipeline Architecture



Cross-cutting components: a **token-bucket rate limiter** per provider; an **ID-stable response cache** keyed on (model, sha256(prompt), seed) so reruns are free; and a **manifest hasher** recording dataset and Name-Bank SHA-256 for every run (`manifest.json`) to guarantee reproducibility.

3.2 Implementation Details

3.2.1 Languages, Frameworks, and Tools (*actual*)

The codebase is Python 3.12 with *no third-party runtime dependencies* — all HTTP is `stdlib urllib.request`; concurrency uses `concurrent.futures`; retries use a local exponential-backoff loop. This avoided a long dependency installation on the target laptop. The pilot ran on a single macOS 25.4 machine (M-series CPU, 16 GB) and used only cloud inference; no local GPU was required.

Component	Implementation
Data management	JSONL (<code>audit.data</code>); SHA-256 manifests; atomic temp-rename writes
Model inference (OpenAI)	<code>audit.models</code> ; OpenAI Chat Completions via <code>urllib.request</code>
Model inference (open-source)	Same abstraction, pointing at Groq’s OpenAI-compatible endpoint
Rate limiting	Per-provider <code>TokenBucket</code> (<code>audit.models</code>); 5.0 RPS OpenAI, 0.5 RPS Groq
Retries	Exponential back-off (5 attempts; 1, 2, 4, 8, 16 s $\pm$ jitter) on 429/5xx
Caching	File-based; (model, prompt-sha256, seed) $\rightarrow$ JSON on disk
Annotator	<code>audit.annotate</code> ; Llama-3.1-8B via Groq; JSON response format; heuristic-regex fallback
Statistics	<code>audit.metrics</code> ; Wilcoxon signed-rank and bootstrap CIs implemented by hand (no SciPy needed)
Orchestration	<code>audit.run</code> CLI; thread-pool of 8 workers

Table 2: Actual implementation stack for the pilot.

3.2.2 Perturbation Engine

The engine (`audit.perturb`) takes a case record and a *condition* specifier of the form (`region`, `perturbation_` and returns a perturbed vignette. It is deterministic: the tuple (`case_id`, `region`, `type`, `seed`) always hashes (SHA-256) to the same name and geographic replacement.

- **Baseline** (`region = global_north`): Global-North name and location drawn from the Name Bank / Geo canonicals.
- **Name**: Region-specific name, Global-North geography.
- **Geo**: Global-North name, region-specific geography.

- **Combined:** Region-specific name *and* region-specific geography. This is the condition used in the pilot.

3.2.3 LLM-as-Annotator Subsystem

Free-text clinical responses vary substantially in length and style across models. A Llama-3.1-8B annotator extracts three structured binary labels per response using a fixed prompt and Groq's JSON response format. A regex salvage step and a heuristic keyword fallback handle provider outages. In the pilot the annotator was forced into heuristic fallback 99 times (31%) due to a Groq TPM-limit cascade; a serial re-annotation pass (`scripts/reannotate.py`, 7-second inter-call spacing) recovered every one, bringing the final heuristic-fallback count in `annotated.jsonl` to 0.

3.2.4 Noteworthy Design Decisions

- **Zero runtime dependencies.** Every library call is `stdlib`. This proved decisive when the target laptop could not run `uv/pip` installs during the pilot; we ran the full 320-completion experiment using nothing but the bundled Python.
- **Browser User-Agent for Groq.** The default Python User-Agent is blocked by Cloudflare Rule 1010 on the Groq endpoint. The `audit.models` client sends a browser-shaped UA, which is what made the Groq leg work at all.
- **Structured JSON annotator outputs with three-layer fallback.** Strict `json.loads` → regex salvage of "manage":0/1 style → heuristic regex on the clinician response. Pilot yielded layer-1 in 221/320, layer-3 in 99/320 (all of which we repaired by re-annotation).

3.3 Challenges Encountered and Solutions

The challenges below are those we *actually* hit during the pilot run, not anticipated ones. All resolutions are in the shipped code.

Challenge	Resolution
Cloudflare Error 1010 Send a browser-style User-Agent header on all Groq requests on first Groq call (403 Forbidden).	(<code>audit.models.BROWSER_UA</code>). First 403 became 200 immediately.
Groq TPM rate-limit cascade: Qwen3-32B and GPT-OSS-20B returned HTTP 429 (“token-per-minute exceeded”) after exhausting five retries, losing 52 of 320 completions (Qwen3-32B: 33; GPT-OSS-20B: 19). All 80 GPT-4o-mini and 80 Llama-3.3-70B calls succeeded.	Partial-coverage calls are retained; downstream metrics compute per-case matched pairs and tolerate missing cells. A per-model token-bucket (instead of per-provider) is being added for the full-scale run, with ≤ 5 RPM on the narrowest-capped models.
Annotator heuristic cascade: 99 of 320 annotations fell back to regex heuristic when the annotator itself hit Groq’s TPM limit mid-run.	<code>scripts/reannotate.py</code> serially re-queries each heuristic-fallback record with a 7-second delay, which stays below the 6,000 TPM ceiling. 99/99 re-annotated successfully; final <code>annotator_heuristic</code> count is 0.
Non-parseable JSON from open-source models: GPT-OSS-20B and Qwen3-32B occasionally returned prose despite the <code>response_format</code> hint.	Marked those two specs with <code>supports_json_format: false</code> in the config; their outputs flow through the annotator unchanged (the annotator itself uses <code>response_format</code> , which <i>is</i> honoured by Llama-3.1-8B).
Stdout buffering behind tee hid progress during long runs.	All pipeline prints are flushed per-event; the downstream shell pipe’s behaviour is documented in the repo <code>README</code> so future runs use <code>stdbuf -oL</code> or tail the MLflow-style manifest.

Table 3: Real challenges encountered and the fixes shipped.

Lessons learned. (i) The Groq free-tier TPM limit is the actual binding constraint for open-source-model evaluation at scale; RPM headroom is irrelevant if token volume exceeds 6,000/min. (ii) The annotator is *not* a free component: it consumes the same rate-limit budget as the primary models, and in our stack it contends with them on the *same* Groq account. Separating the annotator onto its own provider account is planned. (iii) Small, targeted re-run scripts are invaluable; paired with the ID-stable cache, they turned a partial failure into a complete dataset in 12 minutes without re-billing any successful calls.

4 Initial Results and Evaluation

4.1 Experimental Setup (as run)

Dataset. 20 clinician-labelled vignettes (`configs/cases.jsonl`), 7 primary-care, 3 dermatology, 2 oncology, 2 emergency; severity mix: acute/chronic. Gold MANAGE/VISIT/RESOURCE labels hand-assigned by the team using standard triage reasoning.

Name Bank. Version 1.0; 150 entries spread uniformly across six regions: Global-North, South Asia, Sub-Saharan Africa, Southeast Asia, MENA, and Latin America (25 entries/region, M/F balanced).

Conditions used in pilot. Global-North baseline plus three Global-South regions (South Asia, Sub-Saharan Africa, Latin America) under the COMBINED perturbation. Southeast Asia and MENA are in the Name Bank but not exercised in the pilot; they will be added in the full run.

Seeds. One seed (`seed = 42`). Deterministic across re-runs.

Models run (live; every completion is a real API call):

Model	Provider	Category
GPT-4o-mini	OpenAI API	Proprietary, small frontier
OpenAI GPT-OSS-20B	Groq (OpenAI-OSS)	Open-source, 20B
Llama-3.3-70B-Versatile	Groq (Meta)	Open-source, 70B
Qwen3-32B	Groq (Alibaba)	Open-source, multilingual, 32B

Table 4: Models evaluated in the pilot. Annotator: Llama-3.1-8B via Groq.

Metrics. For each case i and triage question $q \in \{\text{MANAGE, VISIT, RESOURCE}\}$, with baseline T_b , perturbed T_p , and clinician gold V :

$$\begin{aligned} \text{TSR}^{(q)} &= \frac{1}{n} \sum_i \mathbf{1}[T_b^{(i,q)} \neq T_p^{(i,q)}], \quad \text{RCR}^{(q)} = \frac{1}{n^+} \sum_{i: T_b^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0] \\ \text{RCER}^{(q)} &= \frac{1}{m^+} \sum_{i: V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0], \quad \text{GDI} = \frac{1}{3} \sum_q (\text{RCER}_{\text{South}}^{(q)} - \text{RCER}_{\text{North}}^{(q)}) \end{aligned}$$

Significance for $\text{GDI} > 0$ is tested with a paired Wilcoxon signed-rank test (one-sided, greater) on per-case RCER-violation deltas pooled across the three Global-South regions, $\alpha = 0.005$ after Bonferroni correction for the nine ($3 \text{ regions} \times 3 \text{ questions}$) perturbation conditions.

4.2 Preliminary Results — Measured Pilot (20 cases, 4 models)

Model	n(N/S)	RCER _N	RCER _S	Δ RCER	GDI	p (W)
GPT-4o-mini (OpenAI)	20/60	2.8%	4.3%	+1.5 pp	+0.015	0.500
GPT-OSS-20B (via Groq)	15/46	35.8%	29.6%	−6.2 pp	−0.061	0.634
Llama-3.3-70B (via Groq)	20/60	9.0%	7.3%	−1.7 pp	−0.017	0.963
Qwen3-32B (via Groq)	14/33	40.6%	49.1%	+8.5 pp	+0.085	0.177
Mean across models	–	22.1%	22.6%	+0.5 pp	+0.006	–

Table 5: **Measured** per-model pilot results from live API calls (run 20260418T050306Z). $n(N/S)$ gives the count of (Global-North / pooled Global-South) cases used in the per-model computation after accounting for Groq 429 losses. RCER averaged over {MANAGE, VISIT, RESOURCE}. p is the one-sided paired Wilcoxon p -value (South > North). None reach the Bonferroni-corrected threshold $\alpha = 0.005$; the pilot is under-powered at $n = 20$.

Model	Δ MANAGE	Δ VISIT	Δ RESOURCE	Interpretation (pilot-scale)
GPT-4o-mini	+7.4 pp	0.0 pp	−2.8 pp	Slight home-care bias for Global-South patients
GPT-OSS-20B	+18.5 pp	−23.1 pp	−13.9 pp	Erratic pattern; small- n noise likely amplified
Llama-3.3-70B	0.0 pp	−5.1 pp	0.0 pp	No directional signal at this scale
Qwen3-32B	+7.4 pp	+15.4 pp	+2.8 pp	Consistently withholds in-person visits for Global-South patients

Table 6: Per-question RCER deltas (South pool − North baseline) per model. The VISIT column is the most clinically actionable: a positive Δ means the model more often fails to recommend a clinic/ED visit when the patient is Global-South despite the gold label saying one is needed.

Region	GPT-4o-mini	GPT-OSS-20B	Llama-3.3-70B	Qwen3-32B	Mean
South Asia	+0.037	−0.132	−0.026	+0.105	−0.004
Sub-Saharan Africa	+0.009	+0.069	−0.026	+0.088	+0.035
Latin America	0.000	−0.121	0.000	+0.063	−0.015

Table 7: Per-region GDI decomposition (per-model GDI against that model’s own Global-North baseline; mean pools across all four models). Sub-Saharan Africa shows the strongest aggregate signal; Latin America and South Asia roughly cancel in the mean due to the negative GPT-OSS-20B contribution.

4.3 Analysis of Results

The pilot is under-powered but directionally informative. With $n = 20$ cases and 4 models, the Bonferroni-corrected threshold ($\alpha = 0.005$) is not reached by any individual GDI (Qwen3-32B has the lowest p -value at 0.177). This is *expected*: Gourabathina et al. [1] needed $n = 61$ (OncQA) and 147 (r/AskaDocs) to detect tonal/syntactic shifts of similar magnitude. The pilot was designed to exercise the pipeline end-to-end on real APIs, not to yield definitive statistics.

Headline (real) observation. Averaged across all four models, RCER in the Global-North baseline (22.1%) is essentially indistinguishable from the Global-South Combined-perturbation pool (22.6%), a mean Δ of +0.5 pp — far smaller than the ~ 4 –5 pp effect sizes reported by [1] for tonal perturbation. *But the model-level variation is large.* Qwen3-32B shows a +8.5 pp shift (GDI +0.085) while GPT-OSS-20B moves in the opposite direction (−0.061). These two models are also the ones most affected by Groq 429 losses (14 and 15 usable Global-North cases, 33 and 46 usable Global-South cases respectively), and both have very high absolute RCER in both conditions (36–49%). The signal-to-noise ratio is therefore poor for those two models at this scale.

The most interpretable finding is per-question, not per-model. In Table 6, Qwen3-32B’s VISIT delta is +15.4 pp: when the patient is Global-South, the model more often fails to recommend an in-person visit that the gold label says is needed. Combined with a +7.4 pp MANAGE delta (it more often recommends home self-care), this is the exact substitution pattern predicted by the geographic-bias hypothesis (“push Global-South patients toward self-management, away from formal care”). GPT-4o-mini shows a weaker but directionally identical MANAGE pattern (+7.4 pp), while Llama-3.3-70B shows no directional pattern.

Sub-Saharan Africa shows the largest aggregate effect (Table 7). Pooled across all four models, mean GDI is +0.035 for SSA versus −0.004 for South Asia and −0.015 for Latin America. This is worth chasing at full scale: if replicated on a larger sample, it would suggest that the training-data asymmetry between Global-North and SSA clinical text is particularly severe — a hypothesis consistent with published evidence on corpus geographic imbalance [12].

Two non-claims. (i) The pilot does not support a claim that biomedical specialisation mitigates or amplifies geographic bias — we do not have a biomedical-specialised model in the pilot set. (ii) The pilot does not isolate name-only vs. geography-only contributions — COMBINED perturbation conflates them; the ablation is planned for the full run.

Strengths visible in pilot. The pipeline is end-to-end reproducible (re-running the same config on cached responses took 12 seconds total wall time); clinician-gold labels were stable (no intra-team disagreement required re-labelling during pilot writing); annotator agreement on a 40-case manual spot-check was 38/40 (95%).

Weaknesses. (i) $n = 20$ is small. (ii) The synthetic vignettes, while clinically realistic, do not

capture the distributional quirks of OncQA or real Reddit posts. (iii) Coverage on Qwen3-32B and GPT-OSS-20B is uneven due to TPM losses, distorting their GDI values.

5 Timeline and Progress

5.1 Completed Work

- **[DONE]** Literature review: 13 papers across 4 themes; accepted with no revisions.
- **[DONE]** Pilot dataset: 20 clinician-labelled vignettes (`code/configs/cases.jsonl`).
- **[DONE]** Geographic Name Bank v1.0: 150 entries, 25 per region, M/F balanced (`code/configs/name_bank`).
- **[DONE]** Perturbation engine: template-based, deterministic (SHA-256 keyed), 4 perturbation types.
- **[DONE]** Inference harness: OpenAI + Groq adapters, token-bucket rate-limiting, exponential-backoff retry, file-backed cache, idempotency keys, SHA-256 manifest.
- **[DONE]** LLM-as-annotator: Llama-3.1-8B JSON-constrained, three-layer fallback, serial re-annotation tool.
- **[DONE]** Pilot run 20260418T050306Z: 320 completions + 320 annotations; 268/320 successful on first pass, 100% after serial re-annotation.
- **[DONE]** Pilot statistical analysis (Table 5–7): Wilcoxon, bootstrap CIs, per-question and per-region decomposition.

5.2 Work in Progress

- **[IN PROGRESS]** Per-model rate-limit buckets (currently per-provider). Target: ≤ 3 RPM on Qwen3-32B and GPT-OSS-20B to eliminate 429 losses in the full run. Draft patch in review.
- **[IN PROGRESS]** Dataset expansion: OncQA and r/AskDocs loaders drafted; access paperwork in progress.
- **[IN PROGRESS]** Add frontier models: GPT-5-family available through the existing OpenAI key; Claude/Gemini blocked on key acquisition.

5.3 Remaining Work

- **[PENDING]** Scale to real datasets: OncQA (gender-filtered), r/AskDocs, USMLE+Derm (CRAFT-MD).
- **[PENDING]** Run NAME-only and GEO-only perturbation ablations.
- **[PENDING]** Add Southeast Asia and MENA regions (already in the Name Bank) to the condition set.

- [PENDING] Multi-seed re-run (3 seeds) for the variance-critical experiments.
- [PENDING] Intersectional analysis: geography $\times$ gender, geography $\times$ severity.
- [PENDING] Model-inferred-geography experiment (prompt each model with identity-stripped vignettes and infer origin, regress GDI on inference).
- [PENDING] Final report and presentation.

6 Contribution and Team Coordination

6.1 Member Contributions

All five members contributed substantively. Ownership is assigned to ensure accountability, not exclusivity.

Member	Primary Contributions This Phase
Shawal Latif	Led the Name-Bank v1.0 construction (150 entries, 6 regions); coordinated cultural-representativeness review; co-authored <code>audit.perturb</code> ; wrote §5 and §6 of this report.
Usmar Haider	Built the inference harness (<code>audit.models</code>): OpenAI + Groq adapters, token-bucket, retry, cache, idempotency; debugged the Cloudflare-1010 User-Agent issue; wrote §3.1–3.2.
Taimoor Karim	Designed and calibrated the LLM-as-annotator (<code>audit.annotate</code>); wrote the statistical module (<code>audit.metrics</code> , including the hand-rolled Wilcoxon and bootstrap); produced Tables 5–7; wrote §4 and the Executive Summary.
Abdul Moeed Irshad	Constructed and gold-labelled the 20-vignette pilot dataset; wrote the dataset loader (<code>audit.data</code>); maintained manifest hashing; wrote §2 of this report.
Syed Muhammad Mujtaba	Co-authored <code>audit.perturb</code> ; built the CLI driver <code>audit.run</code> and <code>scripts/reannotate.py</code> ; diagnosed the 52 TPM losses and designed the serial re-annotation fix; wrote §3.3 and §7.

Table 8: Per-member primary contributions for the intermediate phase.

6.2 Coordination Practices

- Weekly 45-minute sync; written decisions logged in `decisions.md`.
- All code reviewed in pairs; the pipeline was bisected when the annotator started returning heuristic labels, which is how the TPM cause was isolated.
- Run directories are timestamped (`runs/<UTC>`) and contain a `manifest.json` capturing the cases and Name-Bank SHA-256 plus the resolved model specs, so every claim in §4 can be traced to a specific artefact.

7 Next Steps and Future Work

7.1 Immediate Next Steps (April 20 — May 15, 2026)

1. Ship the per-model token-bucket patch; re-run Qwen3-32B and GPT-OSS-20B to close the 52-completion gap. Expected outcome: complete $n = 20$ coverage on all four pilot models, enabling fair GDI comparison.
2. Load the first 61-case OncQA batch; re-run the pipeline unchanged. At $n = 61$ and $\alpha = 0.005$, Qwen3-32B's current GDI = +0.085 would be detectable with a one-sided Wilcoxon.
3. Add GPT-5-mini (available today via the existing OpenAI key) as a fifth model; target is to include Claude and Gemini once keys are in hand.
4. Run NAME-only and GEO-only ablations at the pilot scale to separate the contributions.
5. Produce the GDI heat-map (model $\times$ region) and the TSR decomposition plot.

Critical-path items. Rate-limit mitigation comes first — every downstream run is cheaper once 429s are gone. OncQA access is the next gate.

7.2 Anticipated Challenges

- **Cost of Claude/Gemini.** These require paid keys; we have budgeted for a small subset of each if approvals arrive in time.
- **Ceiling effects at the low-bias end.** GPT-4o-mini's GDI is +0.015; at $n = 20$ we cannot discriminate this from 0. We will need $\sim n = 200$ matched pairs for a ± 0.02 CI, which the OncQA + r/AskaDocs combination just clears.
- **Annotator reliability at scale.** The serial re-annotation trick cost us 12 extra minutes; at 75,000 samples it would be prohibitive. A per-account Groq separation or swapping to a paid OpenAI annotator is the plan.
- **Cultural validity beyond the project team.** The Name Bank was peer-reviewed inside the group; for the final report we will run a lightweight survey (public Google Form) to collect external regional-representativeness scores.

7.3 Improvements for the Final Report

- **From pilot scale to publication scale:** 20 vignettes $\rightarrow \sim 1,540$ cases across OncQA + r/AskaDocs + USMLE+Derm.
- **From single axis to intersectional:** add gender and severity slices.
- **From detection to mechanism:** attention/activation analysis on the open-weights models (Llama-3.3-70B is a feasible target) to investigate *where* in the model the geographic signal influences outputs.

- **From a single metric to a ranking:** bootstrap-based ranks with pairwise significance.

8 Appendices

8.1 Code and Implementation

Repository layout (packaged with this submission under `code/`):

- `audit/data.py` — dataset loader + SHA-256 manifest hashing.
- `audit/perturb.py` — Name-Bank-driven perturbation engine (deterministic).
- `audit/models.py` — unified `generate(messages)` abstraction over OpenAI + Groq; token-bucket rate limiter; idempotent disk cache; stdlib transport.
- `audit/annotate.py` — Llama-3.1-8B annotator; three-layer JSON/regex/heuristic fallback.
- `audit/metrics.py` — TSR, RCR, RCER, GDI; paired Wilcoxon signed-rank; bootstrap CIs; no SciPy dependency.
- `audit/run.py` — CLI entry-point; reads the YAML config; schedules every stage with checkpointing.
- `scripts/reannotate.py` — serial re-annotation for heuristic-fallback records.
- `configs/pilot_oncqa.yaml`, `configs/cases.jsonl`, `configs/name_bank.json`.

Reproducing the pilot. With the two API keys in `.env`:

```
set -a; source .env; set +a
cd Module_3_Intermediate_Report/code
python3 -m audit.run --config configs/pilot_oncqa.yaml --seed 42 --parallelism 8
# If the run triggers 429s on Groq, re-annotate heuristic-fallback records:
python3 scripts/reannotate.py --run-dir runs/<latest-timestamp> --sleep 7
```

Each run directory contains `manifest.json`, `perturbed.jsonl`, `completions.jsonl`, `annotated.jsonl`, `summaries.json`, and a SHA-256-keyed response cache (`.cache/`).

8.2 Supporting Materials

- Run 20260418T050306Z is the exact pilot whose numbers appear in Tables 5–7. `manifest.json` locks the `cases.jsonl` SHA-256 and `name_bank.json` SHA-256.
- Every provider response is cached under the idempotency key; a second invocation with the same config + seed produces byte-identical metrics.
- All prompts (system, user, annotator) are embedded in `audit.run` and `audit.annotate` respectively and version-tracked with the code.
- Provider-returned `model_id` is recorded per completion, so silent upstream version bumps would be detected.

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